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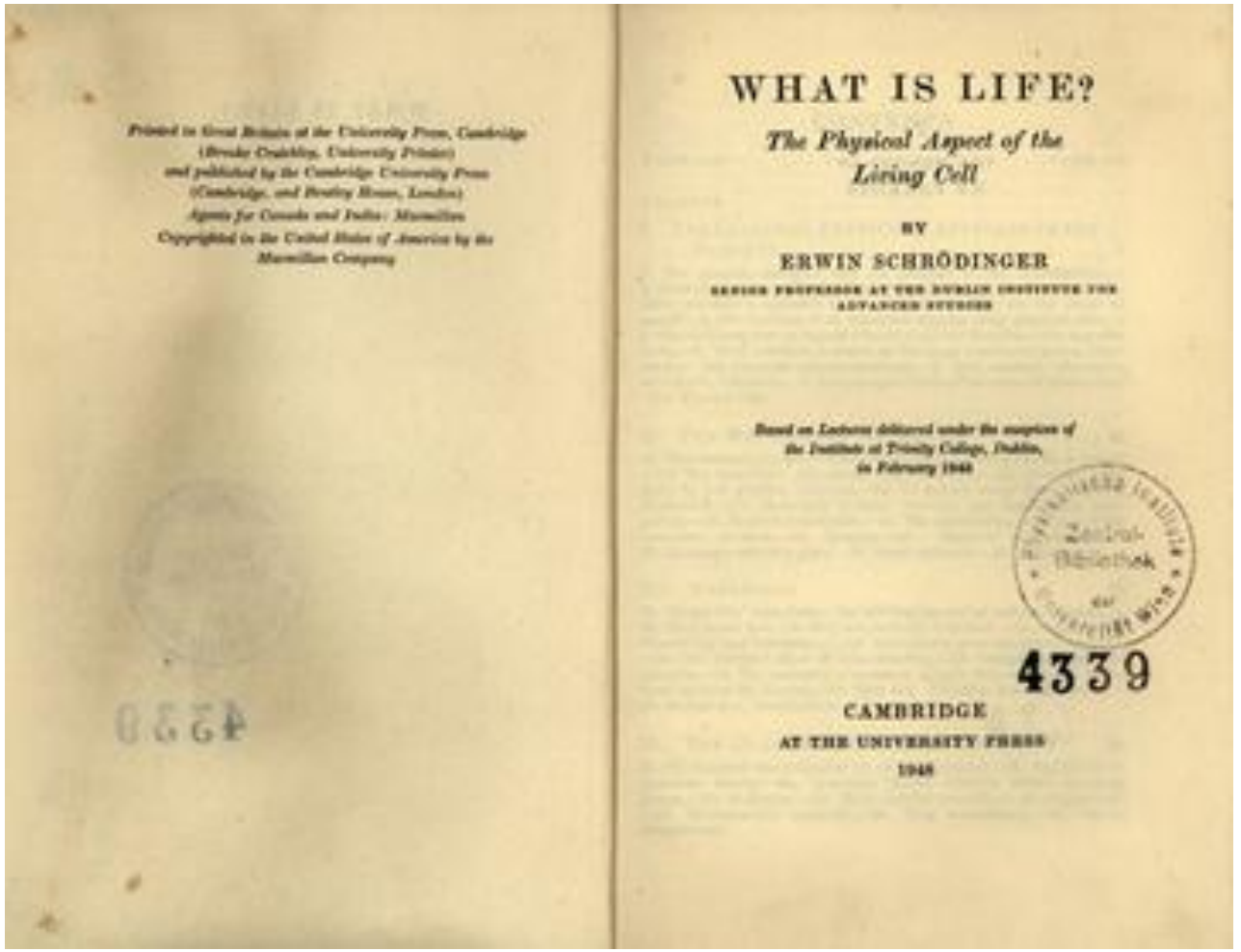
INDIAN INSTITUTE OF TECHNOLOGY DELHI
ABU DHABI

Darwinian Evolution and Molecular Perspective – Origin of life

Lecture 2

Prof. Saurabh Raj
Kusuma School of Biological Sciences
IIT Delhi

What is life?



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WHAT IS LIFE?

The Physical Aspect of the Living Cell

BY

ERWIN SCHRÖDINGER

SENIOR PROFESSOR AT THE DUBLIN INSTITUTE FOR ADVANCED STUDIES

Based on Lectures delivered under the auspices of the Institute at Trinity College, Dublin, in February 1943

4339

CAMBRIDGE
AT THE UNIVERSITY PRESS
1945

What is life?

Living organisms have an ability to:

- Replicate
- Reproduce

What Origin of life deals with:

- How life originated on earth
- How different kind of organisms are found all over the earth

Was life supposed to evolve?

- What we normally think of as life is based on chains of carbon atoms with a few other primary atoms such as oxygen, hydrogen, nitrogen, phosphorous and Sulfur.
- What about silicon-based life?
  - Yes, maybe! but carbon-based seems the most favorable because it has the richest chemistry

***The Anthropic Principle**:*

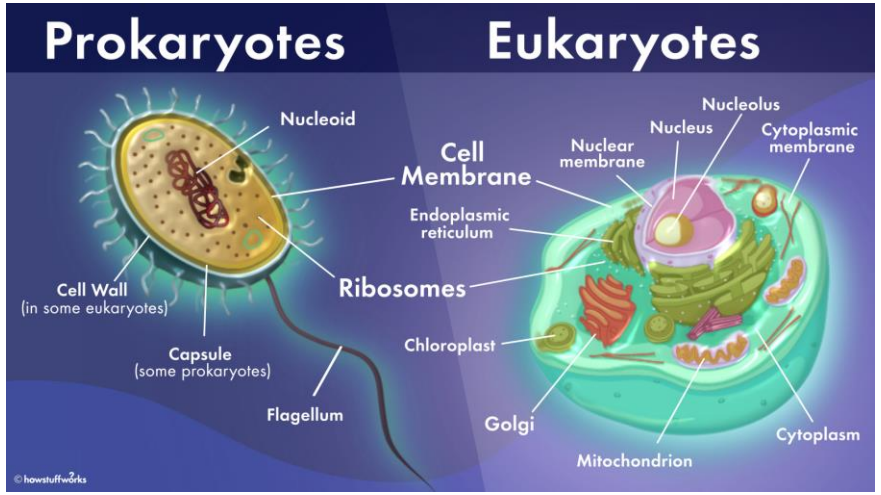
For a given universe, it is possible that the values of the physical constants, will allow the existence of objects like carbon atoms that can act as the building blocks of living systems

https://www.youtube.com/watch?v=KtRcAuunEMg

Understanding the origin of life

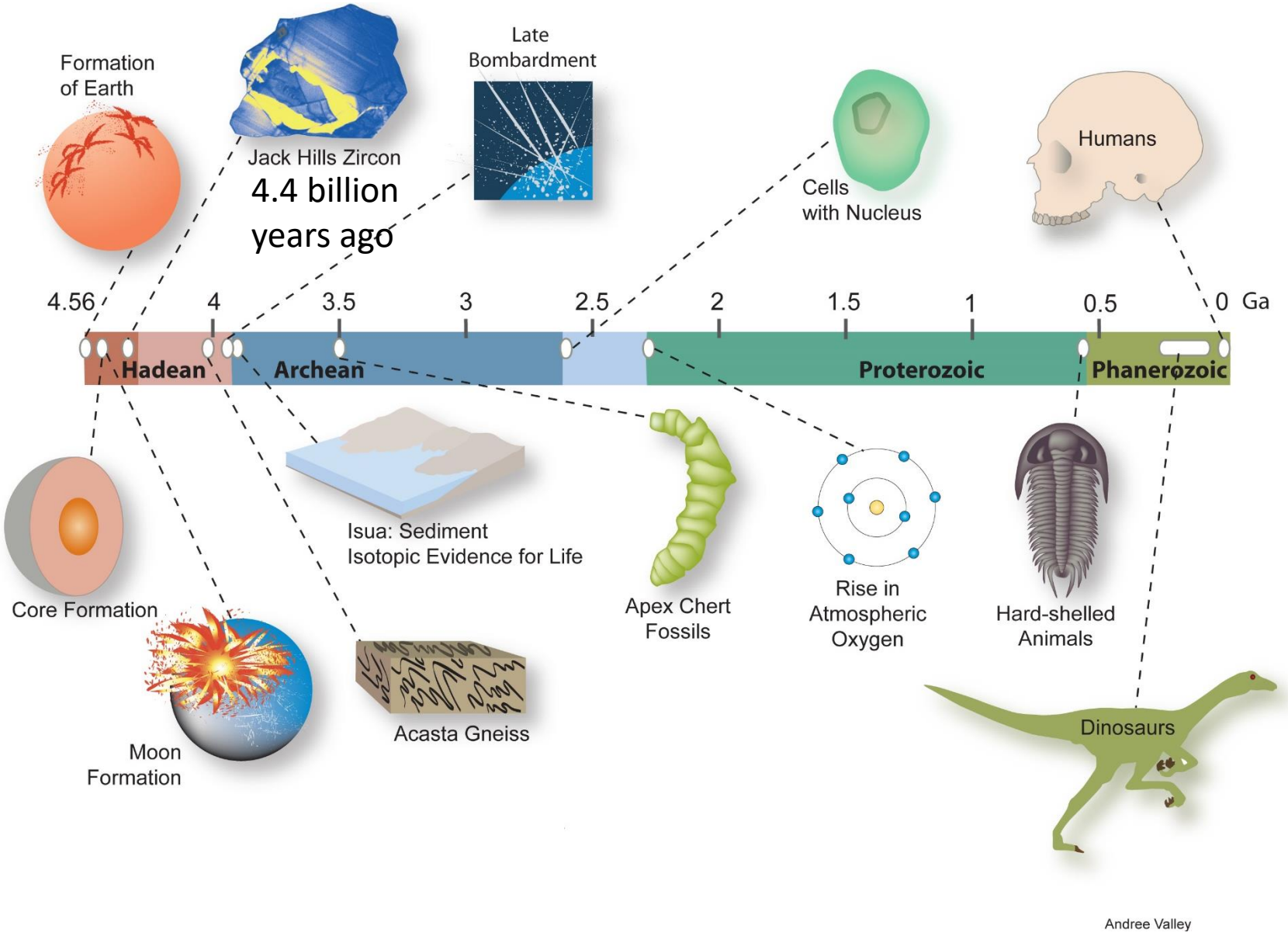
Billion Years	
4.5	Earth Formed
4.2	A stable hydrosphere established
4.2-4.0	Prebiotic world
4.0	Pre-RNA world
3.8	RNA world
3.5	First *prokaryotes* evolve
1.5	First eukaryotes evolve
0.5	Multi-cellular organisms emerge





https://science.howstuffworks.com/life/cellular-microscopic/prokaryotic-vs-eukaryotic-cells.htm

Timeline of early earth events



https://news.wisc.edu/oldest-bit-of-crust-firms-up-idea-of-a-cool-early-earth/

Timeline of early earth events

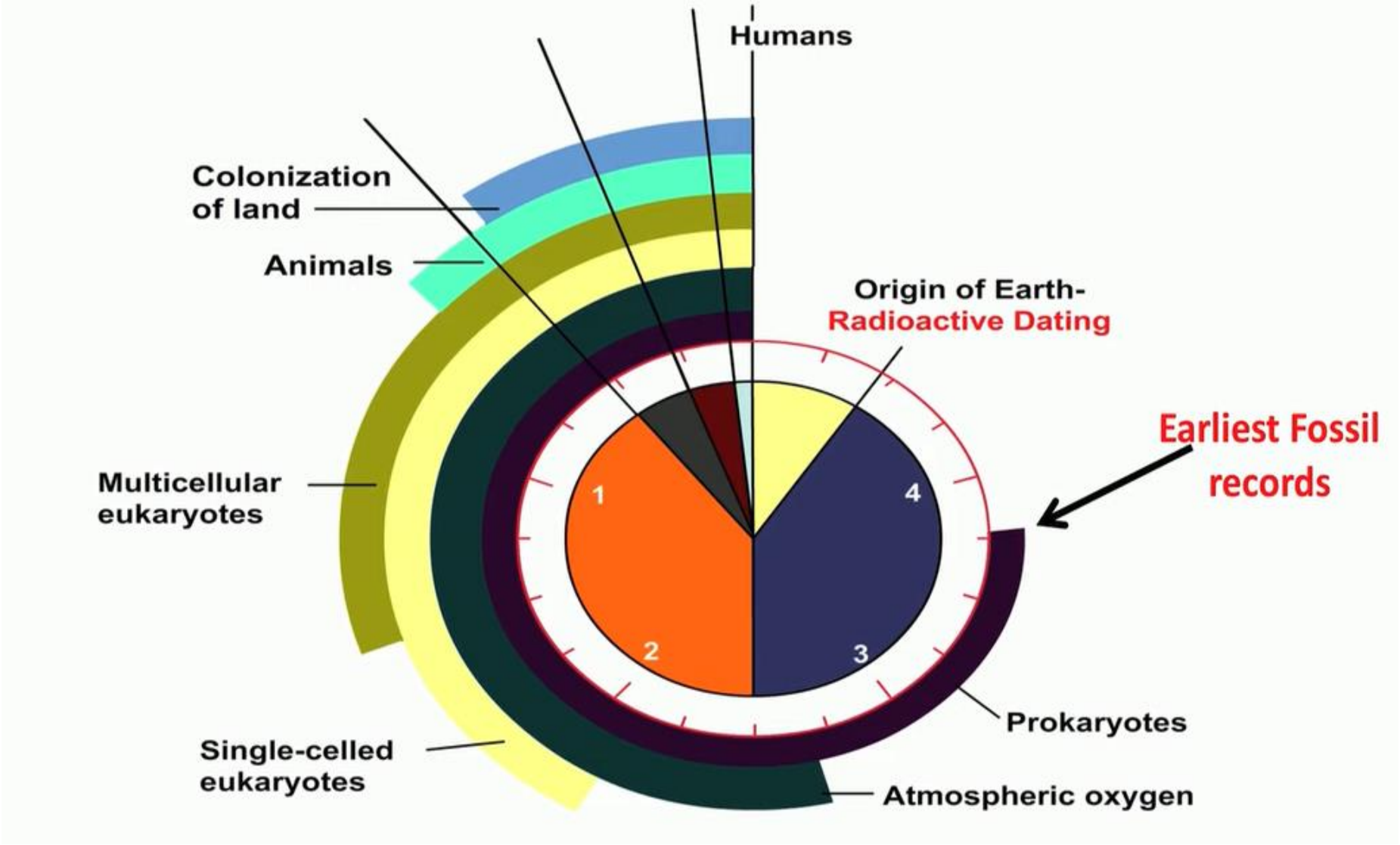


Image from *Campbell Biology*

Understanding the origin of life

This is a ***very hard*** problem, because:

- Organisms are very complex

The interior of cells is over-crowded

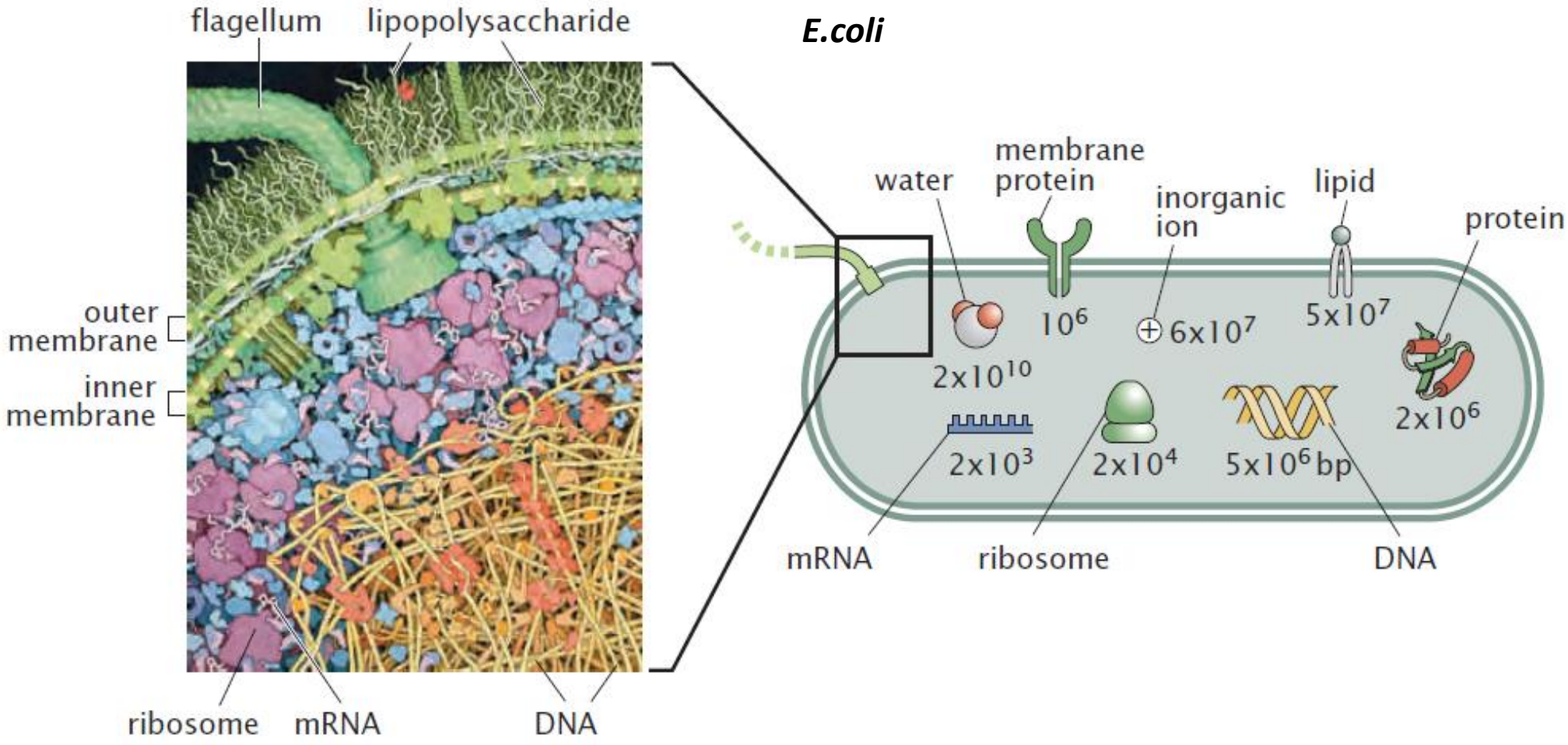
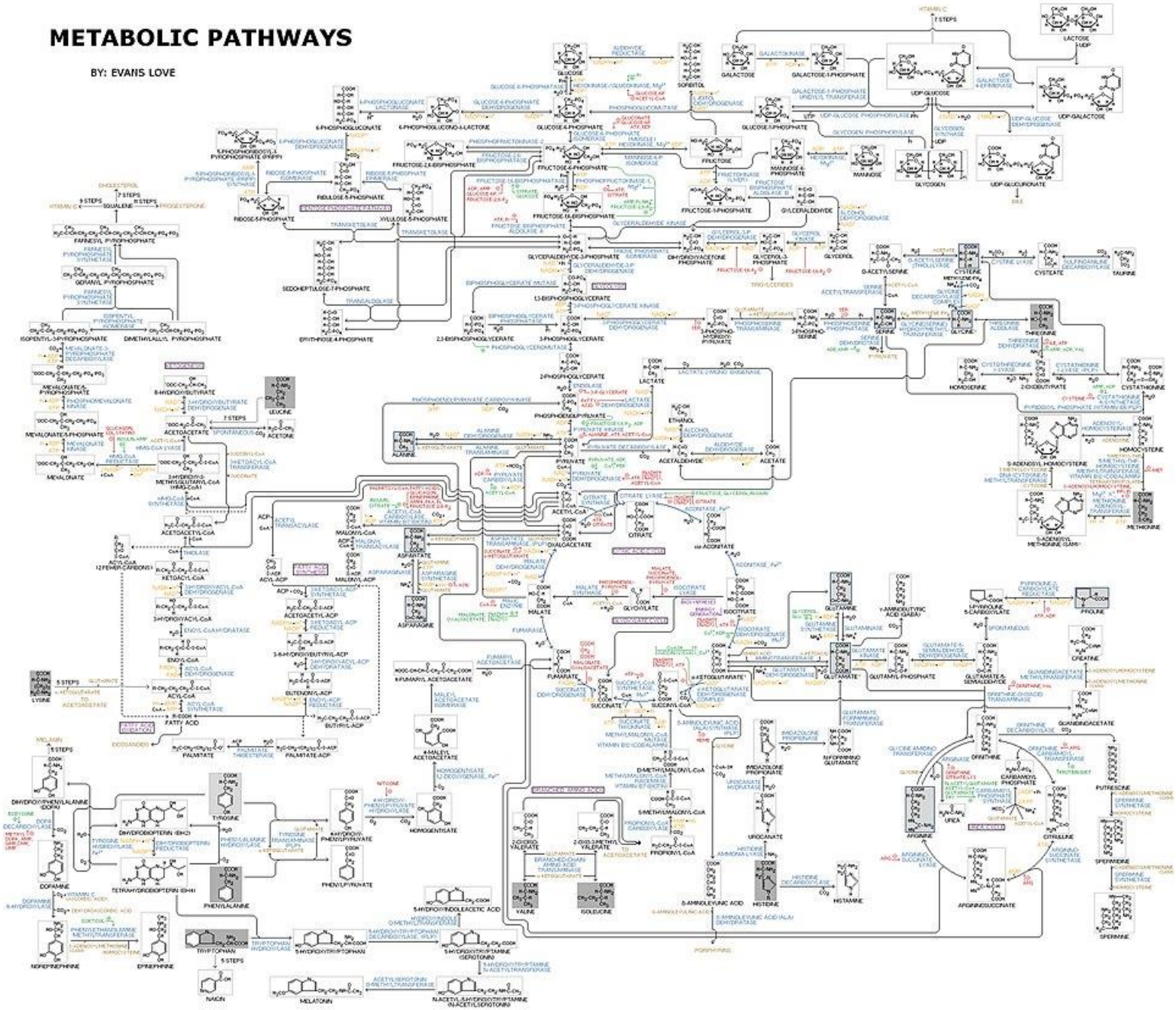


Image from *Physical Biology of Cell*



https://commons.wikimedia.org/wiki/File:Human_Metabolism_-_Pathways.jpg

Understanding the origin of life

This is a ***very hard*** problem, because:

- Organisms are very complex
- We have data from one origin
- It all happened very long time ago – 4.6-4.8 billion years. Humans appeared on earth ~300,000 years ago!
  - ➢We don't have much data
- The condition on earth were predominantly different from what they are now
  - ➢There was no free oxygen

Development of life

- **Chemical Evolution**: Geological molecules to complex organic polymers
- **Acquisition of Replicative ability**: Transition from lifeless chemically interacting entities to self replicating systems
- **Biological Evolution**: Eukaryotes, multi-cellularity, plants and animals, metabolic evolution

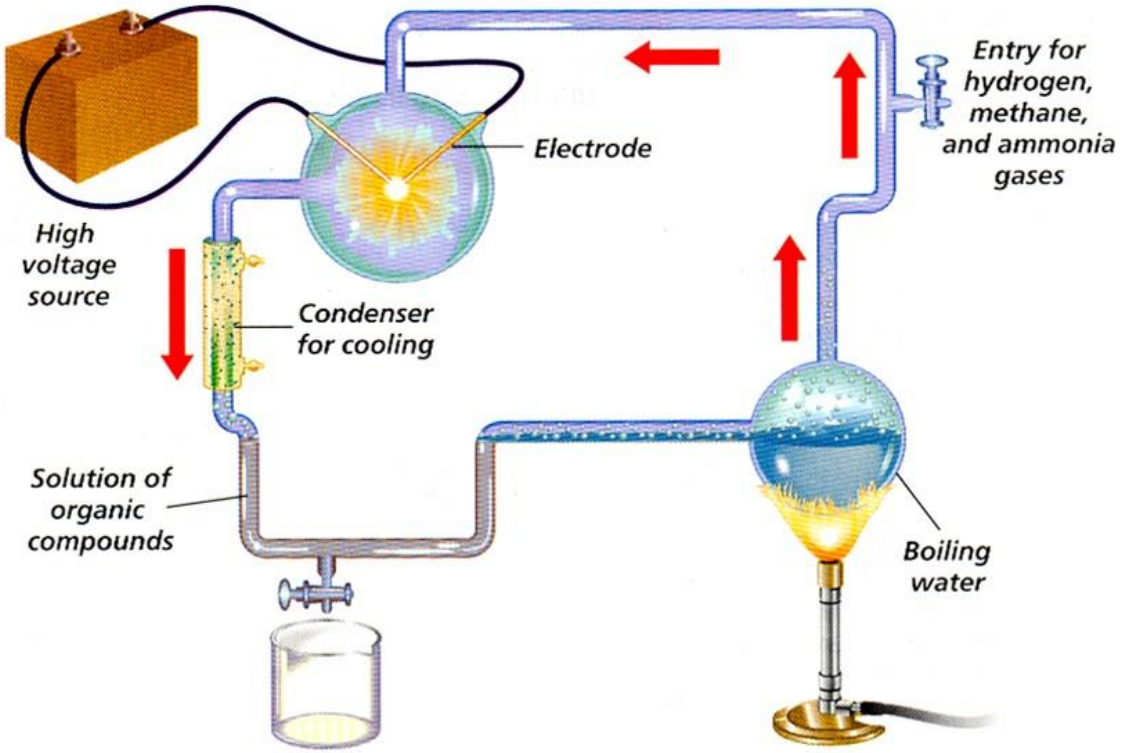
Various hypothesis and theories of life's origin

- 96% of mass of most cells constitute of Carbon, Hydrogen, Oxygen and Nitrogen (CHON) or CHNOPS if one consider Phosphorous and Sulfur as well.
- Chemical properties of carbon
- Primodial earth's atmosphere was reducing unlike present days (N_2, CO_2, CH_4, NH_3, SO_2, H_2O, H_2 and CO)

Hypothesis 1: *Abiotic synthesis of small molecules* (Alexander Oparin and J.B.S. Halden, 1929)

- Any initial biological molecules, such as proteins or sugar, would have been made by abiotic interactions between small molecules.
- High energy discharges (UV/lightening) in the reducing environment of primodial earth would have favored spontaneous synthesis of simple organic molecules from existing geological molecules on earth's surface.

Urey Miller Experiment – a recipe for Primodial Soup



Glencoe Biology, The Dynamics of Life, 2002, p. 350.

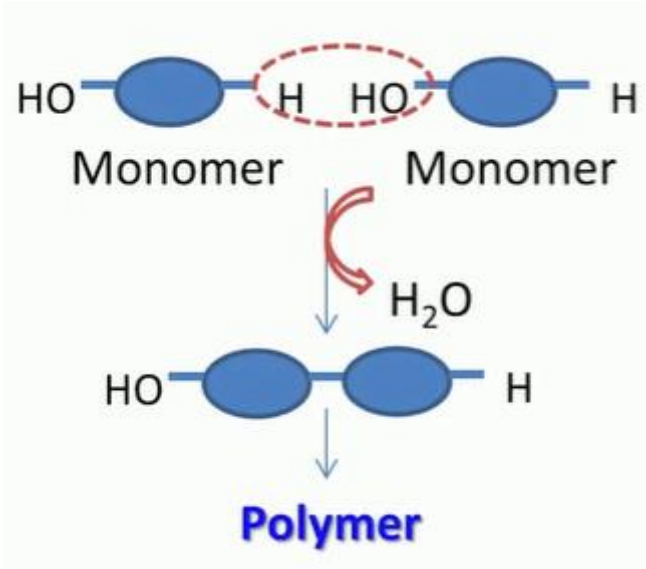
- In 1953 Stanley Miller and Harold Urey performed the first experiment that produced amino acids in what was assumed to be a pre-life atmosphere. They passed a mixture of water vapor, methane, hydrogen and ammonia gases through an electric arc to simulate what would happen if these gases were subjected to lightning.

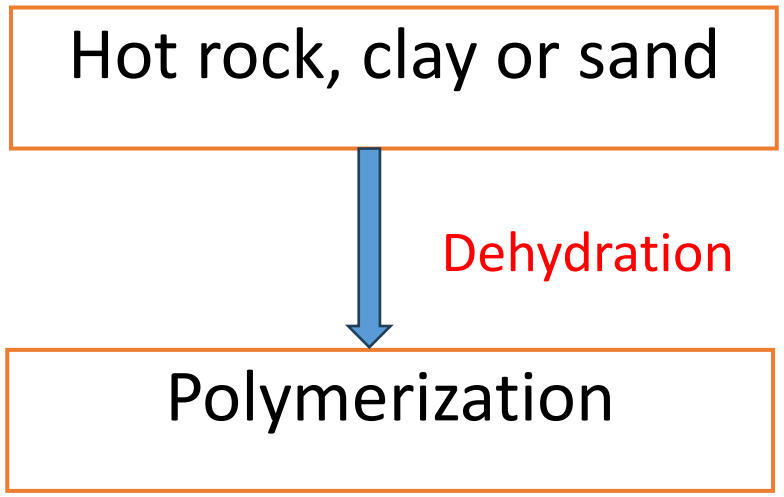
Result:

- **Amino Acids**
- **Sugars**
- **Lipids**
- **Nucleic Acids**
- **Organic Monomers**

But living organisms are much more than monomers!

Hypothesis 2 – Abiotic polymerization of small molecules





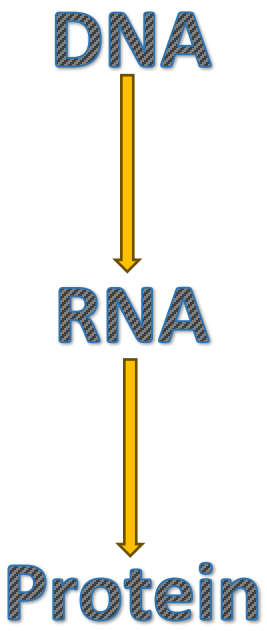
Rate of polymerization >> rate of hydrolysis

No Enzymes

Proof: Sydney Fox

- Dripping organic monomers on hot clay (iron pyrite) or sand
- Charged site
- Polymerization was observed

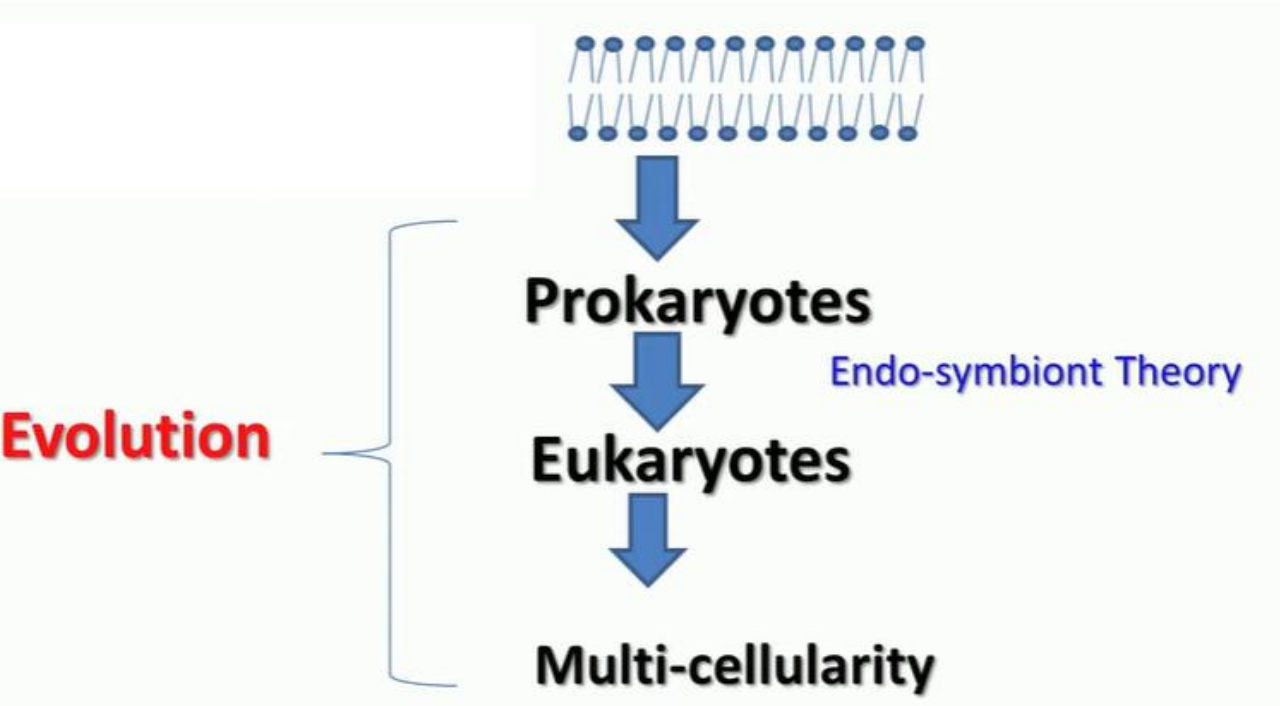
Hypothesis 3 – Abiotic replication



Central Dogma

- Earliest genetic material: RNA
- 1980, Sydney Altman and Tom Cech won Noble prize for demonstrating RNA can act as an enzyme as well.
- RNA can cleave as well as polymerize itself
- Can act as an enzyme itself and is not dependent on proteins
- RNA is used in multiple steps of protein synthesis
- Even Ribosomes are 2/3rd RNA

Hypothesis 4 – Protobionts



- The first formed cells are called probionts
- They were considered as the aggregate of abiotically produced organic molecules surrounded by membranes.
- Through gradual improvements in growth, reproduction, and metabolism, probionts transformed into the first prokaryotic organism.

Hypothesis 1 – Abiotic synthesis of small molecules

Hypothesis 2 – Abiotic polymerization of small molecules

Hypothesis 3 – Abiotic replication

Hypothesis 4 – Protobionts

Metabolic evolution

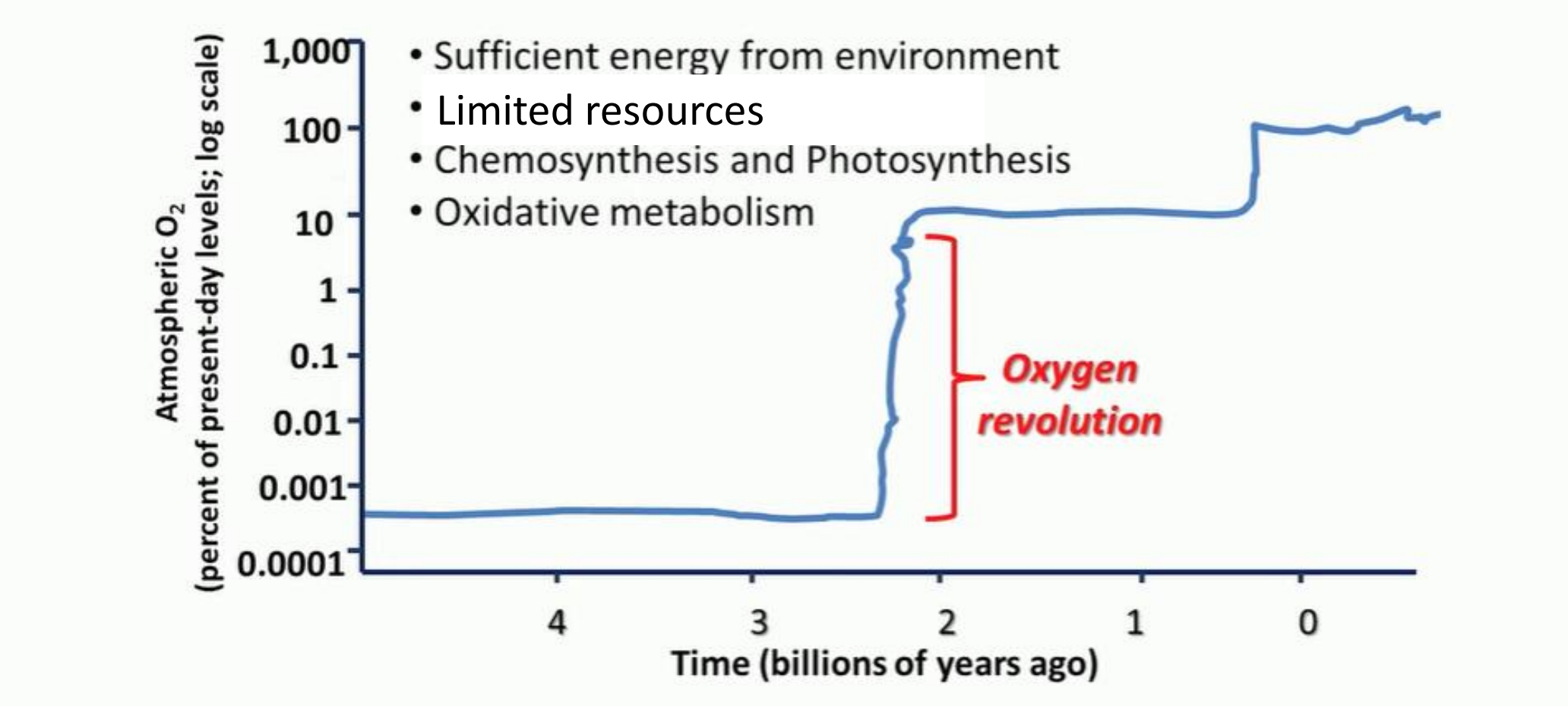
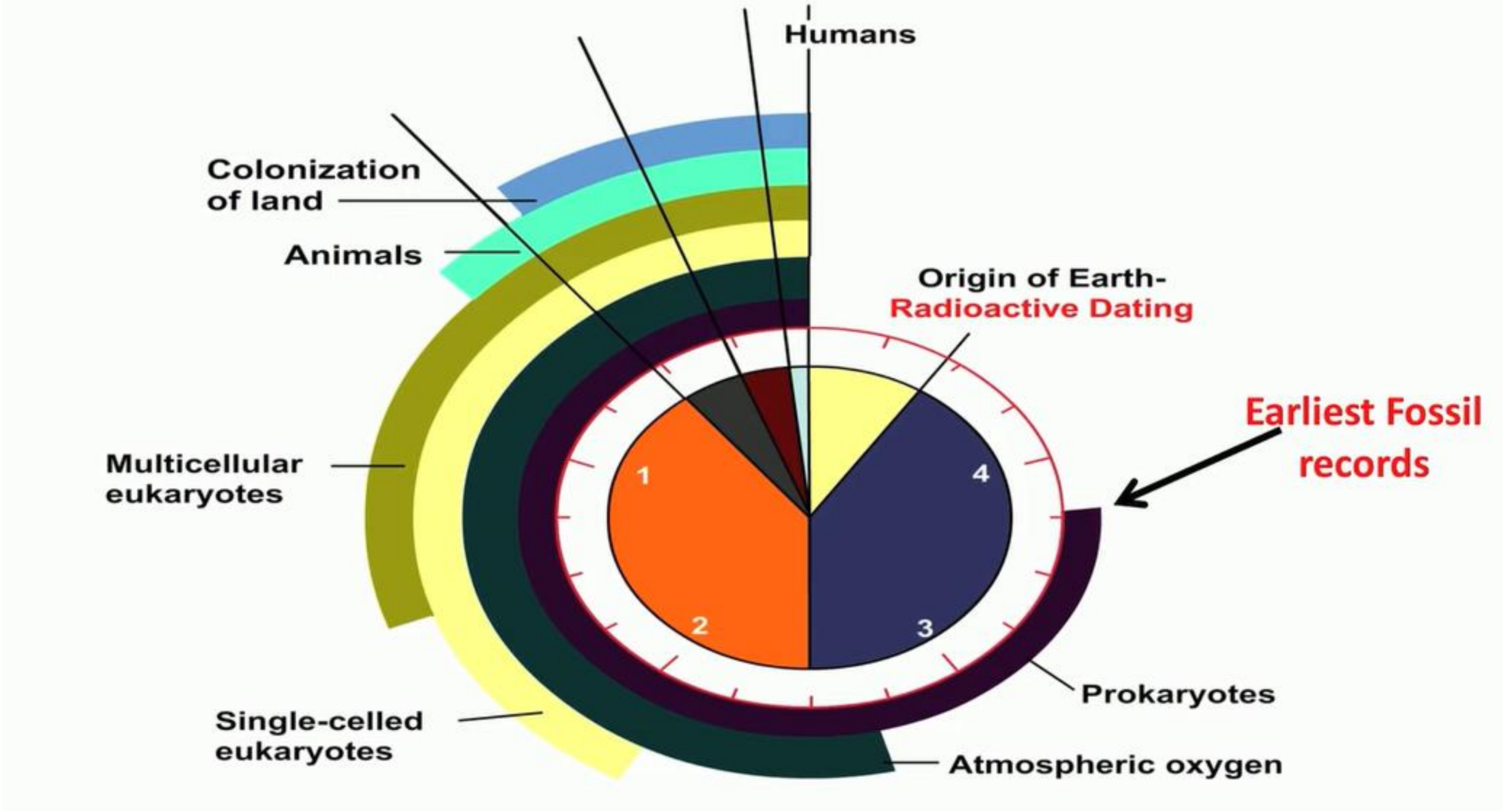
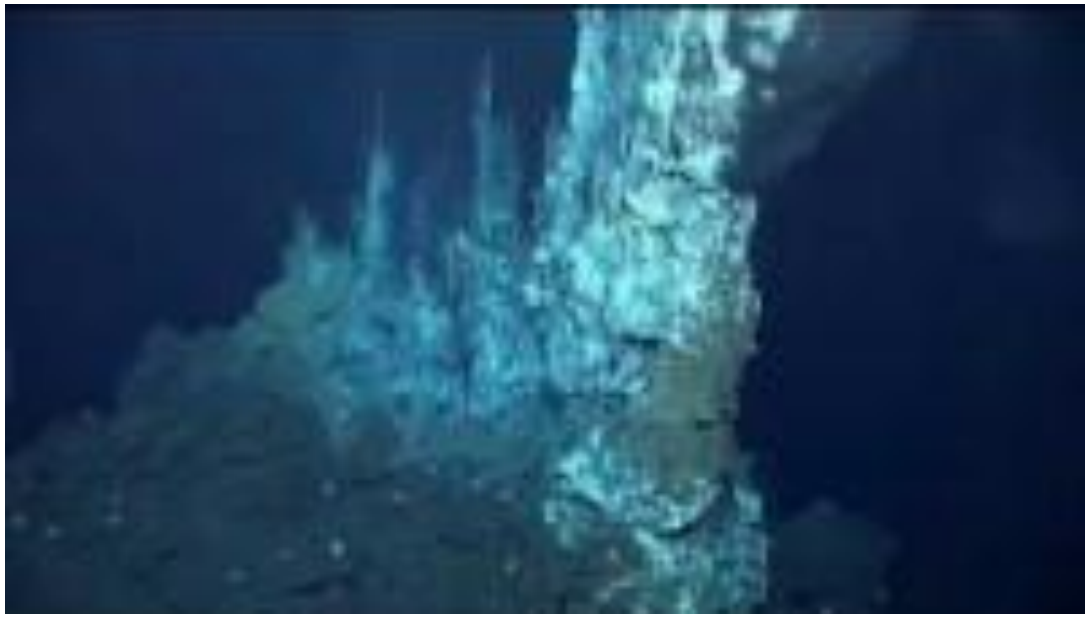


Image from *Campbell Biology*

Timeline of early earth events



How much role does environment play in sustaining life?



Hydrothermal Vents



Thermophiles in Yellowstone national park

https://youtu.be/fZ_x-Yy7sUY?t=7
https://www.columbiatribune.com/story/lifestyle/family/2017/05/24/extremophiles/20804947007/

How much role does environment play in sustaining life?



Cyanobacterium at very low pH



Rio Tinto in Spain, pH 1.7-2.5

http://dx.doi.org/10.1016/j.jaridenv.2013.11.004
https://phys.org/news/2022-02-gain-insights-rio-tinto-microbial.html